GG-H-0080a(DSA-DM)
27 December 1967
INTERIM REVISION OF
Fed. Spec. GG-H-80
18 September 1956

INTERIM FEDERAL SPECIFICATION

HANDLE AND BLADE SET, SURGICAL KNIFE, DETACHABLE BLADE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-H-80.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers surgical knives with detachable blades and handles for various uses in surgery.

1.2 Classification.

1.2.1 Sizes and types. Handles and blades for surgical knives in accordance with this specification shall be furnished in the following sizes and types as specified (see 6.1). Size 1 and size 2 handles are for use with size 1 and size 2 blades, respectively.

1.2.2 Handles.

Size 1. Handles, narrow nose.

Type A.	Minor operating, short.
Type B.	Minor operating, long.
Type C.	Minor operating, offset.
Type D.	Dissecting, short.
Type E.	Dissecting, long.

FSC 6515, 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Size 2. Handles, wide nose.

Type A.	Major operating, short.
Type B.	Major operating, long.
Type C.	Major operating, offset.
Type D.	Dental mechanics.

1.2.3 Blades.

Size 1. Blades, small tang.

Type AA.	Minor operating, small.
Type BB.	Minor operating, large.
Type CC.	Bistoury, straight.
Type DD.	Bistoury, curved.

Size 2. Blades, large tang.

Type AA.	Major operating, small.
Type BB.	Major operating, medium.
Type CC.	Major operating, large.
Type DD.	Major operating, special.
Type EE.	Major operating bistoury, straight.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the currentversion before use.

STANDARDS

Federal

FED-STD-123 Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-P-116 Preservation, Methods of.
MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM B122-60	Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar.
ASTM D700-63T	Phenolic Molding Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

The United States Pharmacopeial Convention, Inc.
The Pharmacopeia of the United States of America (U.S.P)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

Visual Aids for Classification of Defects - Surgical Blades

(Copies of this brochure are available from the contracting officer upon request.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Handles. Handles shall be fabricated of one of the following materials as specified.

3.1.1.1 Corrosion-resistant steel. Corrosion-resistant steel used for handles shall be in accordance with class 6, type 430 of GG-I-526. Handles shall be finished as specified in 3.5.3.

3.1.1.2 Nickel-silver. Nickel-silver used for handles shall be fabricated of grade A or grade B nickel-silver, as specified.

3.1.1.2.1 Grade A. Grade A nickel-silver shall be in accordance with class 8, type 752 of GG-I-526. Handles shall be finished as specified in 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.1.1.2.2 Grade B. Grade B nickel-silver shall be American Society for Testing and Materials (ASTM) B122-60, alloy 3, conforming to chemical composition listed below, and shall be finished as specified in 3.5.3.

Copper.....	63.5 to 68.5 percent
Nickel.....	9 to 11 percent
Zinc.....	19 to 24 percent
Other elements.....	0 to 2 percent

This material shall comply with corrosion-resistance requirements specified in GG-I-526 for class 8, type 752.

3.1.1.3 Plastic. When plastic handle is specified (see 3.5.1.1), the plastic used for the handle proper shall be fabricated of red colored phenolic material, conforming to ASTM D700-63T, type 2SS or 11SS. Finish shall be as specified in 3.5.3.

3.1.1.4 Handle nose. The nose of the handle shall be fabricated of the same material used for the handle proper, except where the handle is plastic, the nose shall be fabricated of corrosion-resisting steel, nickel-silver, or magnetic iron.

3.1.2 Blades. Blades shall be fabricated of a high grade steel having a uniform, fine grain structure. Material shall be either composition A or B, as hereinafter specified.

3.1.2.1 Composition A. Composition A shall be carbon steel with a carbon content of no less than 1.20 percent. Hardness of finished blade shall be Rockwell 30N 75.5 to 83.

3.1.2.2 Composition B. Composition B shall be a high chromium stainless steel with a carbon content of no less than 0.50 percent. Hardness of finished blade shall be Rockwell 30N 74 to 83.

3.2 Marking for identification. Each blade and each handle shall be permanently and legibly marked with the commercial designation number specified in table I, and the name or registered trademark of the manufacturer. Blades fabricated from composition B material shall be marked with the legend: "stainless".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Table I. Commercial Designation Numbers

Handles			Blades		
Size	Type	Commercial designation number	Size	Type	Commercial designation number
1	A	3	1	AA	15
	B	3L		BB	10
	C	3LA		CC	11
	D	9		DD	12
	E	7			
2	A	4	2	AA	20
	B	4L		BB	21
	C	4LA		CC	22
	D	6		DD	23
				EE	25

3.3 Tolerances. Tolerances indicated delineate acceptable extreme dimensions, and are not intended to permit variations within a single instrument which will sacrifice uniformity, symmetry and first quality workmanship. For dimensions shown in the figures and not otherwise limited, tolerances shall be permitted as shown in table II.

Table II. Tolerances for Dimensions of Handles and Blades

Dimension	Tolerance (plus or minus)
0.25 inch or less	0.0156 inch.
Over 0.25 inch to 1 inch	0.0312 inch.
Over 1 inch to 3 inches	0.0625 inch.
Over 3 inches	0.25 inch.
Angles	0.50 degree.

3.4 Gages. Applicable dimensions of the handles and blades shall be within the limits of suitable "Go" and "No Go" gages, the requirements of this specification applying in all other respects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.5 Handles.

3.5.1 Design. Each handle shall consist of the handle proper and the nose. Handles shall conform to the applicable design and dimensions shown in figures (fig.) 1a to 1f and 2a to 2d, inclusive, for the various sizes and types. Handles shall fit accurately, and shall be easily detachable from the blades from which they are designed, and shall be interchangeable with handles of the same size.

3.5.1.1 Dental mechanics handle. Dental mechanics handle shall be plastic with metal nose (see 3.1.1.3). When tested in accordance with 4.4.2.3, the handle proper and the nose shall remain firmly together in proper alignment; the plastic shall show no cracks, become soft or sticky, or otherwise show evidence of deterioration.

3.5.1.2 Other handles. Handles shall be one-piece construction of either corrosion-resisting steel or nickel-silver.

3.5.2 Noses. Noses shall comply with the design and dimensions shown in figure 3 except for nose of size 2, type D, dental mechanics handle for which the G dimension shall be 0.023 to 0.035 inch. Dimensions not shown shall be approximately in proportion with those indicated in the figure.

3.5.3 Finish.

3.5.3.1 Metal handles. Handles shall be free from burs and sharp or rough edges. Corrosion-resisting steel handles shall have a satin finish. Grade A or grade B nickel-silver handles shall have a satin chromium finish.

3.5.3.2 Plastic handles. Plastic handles shall have a smooth finish except for the hand gripping area (as shown in figure 2d) which shall have a textured finish. Nose of plastic handle made of corrosion resisting steel shall have a satin finish. Nose of plastic handles made of nickel-silver shall have a satin chromium or cadmium plated finish. Nose of plastic handles made of magnetic iron shall have a cadmium plating of 0.0001 to 0.0002 inch with a supplementary clear chromate conversion treatment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.6 Blades.

3.6.1 Design. Blades shall be of steel, properly hardened and tempered, and of one-piece construction. Blades shall comply with the applicable design and dimensions shown in figures 4a to 4d and 5a to 5e, inclusive, for the various sizes and types. That part of the blade which is adjacent to the perforation and which fits into the groove of the nose of the handle shall be 0.0145 to 0.0165 inch thick. The remainder of the blade shall be 0.0145 to 0.0212 inch thick, except that it may be thicker along the back edge as a means of reinforcement. Blades shall fit securely, shall be easily detachable from the handles for which they are designed, and shall be interchangeable with blades of the same size.

3.6.2 Tang. The tang shall comprise that part of the knife blade which fits the knife handle. It shall have a perforation and an angular rear edge which shall comply with the design and dimensions shown in figure 6, except, as an alternate, dimension E shall apply only up to a point 0.115 ± 0.010 inch from the proximal end of the perforation, and the remainder of that portion of the perforation (remainder of dimension H) shall be 0.095 to 0.109 inch for size 1 blades and 0.145 to 0.159 inch for size 2 blades. Dimensions not shown shall be in proportion with those indicated in the figure.

3.6.3 Cutting edge and point. The cutting edge of the blade shall be central and uniform, as is consistant with the best manufacturing procedures. (The cutting edge includes the beveled portion of the blade.) The point of the blade shall be well defined and sharp. Points shall not be rounded more than 0.005 inch radius or blunt beyond a 0.005 inch flat, or bent more than 0.010 inch off center. No blade shall contain a nick greater than 0.004 inch. (A nick is defined as a chipped-out, broken-out, indented, or bent-out piece of metal, a semi-circular projection, or any similar gap, indentation or projection.) No blade shall contain more than two nicks of a size greater than 0.002 inch on the cutting edge. Such nicks, if present, shall be no closer than 1/4 inch apart on the cutting edge. No blade shall contain a feather greater than 0.005 inch in length, nor more than two feathers smaller than 0.005 inch in length and no closer than 1/8 inch apart. (A feather is defined as a thin, turned, or curled edge not removed by honing or buffing.) There shall be no burs, jags or other defects

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

on the cutting edge. (A bur is defined as a piece of metal projection not normally inherent to a smooth uniform surface.) (A jag is defined as several small toothlike projections or similar indentations, individually smaller than a nick, and collectively greater than 0.015 inch in length.) The requirements of this paragraph shall be determined as specified in 4.4.3.1.

3.6.3.1 Sharpness and cutting edge angle. The point and the cutting edge shall be surgically sharp. The cutting edge shall have an included angle not greater than 36 degrees. The two angles forming the included angle shall each be 13 ± 5 degrees.

3.6.4 Stiffness. Blades shall have sufficient stiffness not to deflect sideways beyond the torque limits specified under any conditions of normal use. The tang shall have sufficient resistance to lateral deflection to obviate the possibility of accidental separation of the blade from the handle. When the blade is tested in accordance with 4.4.3.3, in no case shall the applied torque deflect the movable arm through an angle of more than 14 degrees for the point and 16 degrees for the tang. As an exception, when size 2, type EE blade is tested in accordance with 4.4.3.3, in no case shall the applied torque deflect the movable arm through an angle of more than 14 degrees for the point or the tang.

3.6.5 Finish. All edges of the blade shall be smooth. Burs (such as manufacturing process burs) shall not exceed 0.001 inch on all edges except the cutting edge, which shall be as specified in 3.6.3 and 3.6.3.1. Finished blades shall be free from pits and grind marks (see 3.6.6.4). The requirements of this paragraph shall be determined as specified in 4.4.3.1.

3.6.6 Assembly. Blades shall be sterile, free from pyrogens, and safe (nontoxic). Each blade shall be packaged in an individual, sealed, aluminum foil package, or in a combination paper-plastic package (see 5.1.3.2). Each sealed package shall maintain the sterility of its contents, and shall be designed so as to permit ready dispensing of the blades by sterile techniques.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

3.6.6.1 Sterility. When tested as specified in 4.4.4, blades shall comply with sterility test requirements of the current edition of The Pharmacopeia of the United States of America (U.S.P.).

3.6.6.2 Freedom from pyrogens. When tested as specified in 4.4.5, blades shall comply with pyrogen test requirements of the current edition of U.S.P.

3.6.6.3 Safety. When tested for determination of acute toxicity, as specified in 4.4.6, there shall be no deaths or discernible symptoms of toxicity of the test animals. In determining tissue reaction (see 4.4.6), the injection sites on the test animals shall be free from evidence of irritation or toxicity. Trauma shall be ruled out with suitable saline control sites.

3.6.6.4 Freedom from corrosion and foreign matter. When tested as specified in 4.4.3.1, the blades shall be free from corrosion, oil, grease, or burs (see 3.6.5). No blade shall contain metal particles clinging to blade surfaces. The presence of pieces of packaging materials, which are integral to the inside of the package, and which cover no more than one percent of the blade surface other than the cutting edge, shall not be cause for rejection. The cutting edge shall be completely free from such materials. Very small, slightly discolored areas, covering no more than one percent of the total blade surface, shall not be cause for rejection.

3.7 Workmanship. Workmanship shall be first class throughout. Blades and handles shall be free from defects which detract from their appearance or may affect their serviceability.

3.8 Certificates.

3.8.1 Certificate of sterility. Unless otherwise specified a certificate of sterility, listing all sterility lot numbers, and stating that samples from each sterility lot have been tested for sterility as required in 3.6.6.1 and 4.4.4 and were found to be sterile, shall be furnished. The certificate shall reference the method used in determining sterility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.8.2 Certificate of freedom from pyrogens. Unless otherwise specified, a certificate of freedom from pyrogens, listing all lot numbers and stating that samples from each lot have been tested for pyrogens as specified in 3.6.6.2 and 4.4.5, and were found to comply with such requirements shall be furnished. The certificate shall reference the method used in determining freedom from pyrogens.

3.8.3 Certificate of safety. A certificate of safety, listing all lot numbers, and stating that samples from each lot have been tested for safety as specified in 3.6.6.3 and 4.4.6, and were found to comply with such requirements shall be furnished. The certificate shall reference the method used in determining safety.

3.9 For military procurements. For military procurements the certificates specified in 3.8 shall be furnished as specified below.

3.9.1 Source inspection. When source inspection is required, certificates of sterility shall be furnished to the procurement activity via the government inspector.

3.9.2 Destination inspection. When destination inspection is required, certificates of sterility shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, ATTN: Quality Control Branch, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 Unit of product. The unit of product for inspection purposes only shall consist of one blade or one handle.

4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection level and acceptable quality level (AQL) shall be as indicated in table III. Unit of product shall be as indicated in 4.2.1.

Table III. Sampling for Examination

Component	Type defect	Inspection level	AQL (percent defective)*
Handle	Major	S-3	1.0
	Minor A	S-3	2.5
	Minor B	S-2	2.5
Blade	Major A	S-4	0.40
	Major B	S-3	1.0
	Minor A	S-1	1.5
	Minor B	S-1	2.5
	Minor C	S-1	4.0

* If acceptance number is not exceeded, lot shall be accepted.

4.2.3 For tests.

4.2.3.1 For sterility test. Sampling for sterility test shall be as indicated for Liquids and Suspensions Under Sterility Tests in the current edition of the U.S.P.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

4.2.3.2 For freedom from pyrogens and safety tests. The number of samples required for the freedom from pyrogens and the safety tests (see 4.4.5 and 4.4.6) shall be taken from each sterility lot for test.

4.2.3.3 For hardness and stiffness. Sampling for hardness and stiffness shall be in accordance with MIL-STD-105, inspection level S-1. The AQL shall be 2.5. Unit of product shall be as specified in 4.2.1.

4.2.3.4 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105 and table IV.

Table IV. Sampling for Other Tests

Component	Characteristic *	Req.	Test procedure	Unit of product	Insp. level	AQL
Handle	Chemical Composition	3.1.1	4.4.2.2	1-Handle	S-1	**
	Boil Test	3.1.1.2	GG-I-526	1-Handle	S-2	**
	Copper Sulfate Test	3.1.1.1	GG-I-526	1-Handle	S-2	**
	Autoclaving	3.1.1.3 3.5.1.1	4.4.2.3	1-Handle	S-2	**
Blade	Chemical Composition	3.1.2	4.4.3.4	1-Blade	S-1	**

* As applicable.

** Acceptance number shall be zero for all sample sizes.

4.2.4 Sample unit usage. Where feasible the same sample unit shall be used for the determination of two or more test characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.3 Inspection procedure. Examination and testing shall be conducted in accordance with Table V, classification of defects, and with the tests in section 4, to determine compliance with all requirements contained in this specification. Inspection is not restricted to the defects listed below. Any sample unit containing one or more defects shall be rejected (defective) and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table V. Classification of Defects in accordance with MIL-STD-105.

HANDLE

Categories	Defects
Critical:	None defined
Major:	
101	Handle not size and type corresponding to stock number required by contract.
102	Nose dimensions except A, B, C, I, and J not within limits set forth in fig. 3 (or fail to pass gages for these dimensions).
103	Handle not free from cracks, laps, folds or distortion.
104	Edges and corners not free from burs and sharpness.
105	Finish not as specified in 3.5.3.
106	Labels or printed marking on unit container not correct (particularly stock number and nomenclature).
Minor A:	
201	Finish not free from marks or excessively scratched surface areas.
202	Labels or printed marking not legible and not free from defacing marks, scratches and tears.
Minor B:	
251	Nose dimensions A, B, C, I, and J not within limits set forth in fig. 3.
252	Applicable handle dimensions shown in figures 1 through 2 not within tolerances specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

BLADE (See limits of acceptance in applicable paragraph of section 3 of this specification.)

Critical: None defined

Major A *
- 101 Point rounded, blunt, bent or broken.
- 102 Cutting edge and point with feathers, nicks, burs, pits, or jags.

Major B:
- 151 Blade not free from corrosion, stains, oil, grease, and individual plastic or paper-plastic packaging material or other foreign matter.
- 152 Blade not of size or type specified.
- 153 Tang not of design and dimension (including compliance with gages) specified.
- 154 Angle of cutting edge not as specified.

Minor A:
- 201 Cutting edge and point not central and uniform.
- 202 Blade edges (except cutting edge) not smooth and not free from pits, burs or grind marks.
- 203 Blade (composition B) not legibly marked "stainless".
- 204 "Dished" (misshaped) tang.
- 205 Burs, chips, pits, or breaks at tang.

Minor B:
- 251 Individual package (composition A) not fitted with anti-corrosive paper.
- 252 (Composition A) - Anticorrosive paper does not adhere to inner surface of individual package.
- 253 Individual package flaps not of size specified.
- 254 Contents not easily removed.
- 255 Individual package and marking not as specified.

Minor C:
- 301 Blade (except tang) dimensions not within tolerance.
- 302 Blade thickness dimension not within limits.
- 303 Intermediate package and marking not as specified.
- 304 Exterior container and marking not as specified.

* Refer to brochure entitled "Visual Aids for Classification of Defects-Surgical Blades" for illustrated guides of defects included in this category. Copy available from contracting officer upon request.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

4.4 Tests. Samples shall be subjected to the following tests:

4.4.1 Compliance with gages. Unless otherwise specified by the procuring agency, the design and dimensions of the nose of the handle and the tang and perforation of the blade shall be checked by use of suitable "Go" and "No Go" gages (fig. 7 and 8).

4.4.2 Tests for handles.

4.4.2.1 Steel handles. Material shall meet resistance to copper sulfate test and resistance to boil test requirements as specified in GG-I-526.

4.4.2.2 Nickel-silver handles. At the request of the procuring agency, the contractor shall furnish millings or drillings of handle stock which shall be used in determining compliance with the chemical requirement specified in 3.1.1. The contractor shall guarantee the sample millings or drillings to be of the same material as the stock from which the particular handle was manufactured. The millings or drillings shall weigh not less than 4 ounces and shall be fine, clean, dry, and free from foreign matter. They shall be taken at slow speed and shall not be oxidized. Unless otherwise specified by the procuring agency, spectrographic analysis may be used in lieu of the chemical analysis test.

4.4.2.3 Plastic handles. Handles made of plastic shall be autoclaved for at least 20 minutes at a temperature of at least 250 degrees Fahrenheit (° F.). Then the handles shall be allowed to cool to room temperature and examination shall be made to determine compliance with the requirements of 3.1.1.3 and 3.5.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.3 Tests for blades.

4.4.3.1 Microscopic inspection. Blades shall be inspected under 10 power magnification using a binocular microscope and direct (onto the top of the stage) illumination by two fifteen-watt fluorescent lamps. The lamps shall be positioned close to the stage to provide the optimum illumination necessary for complete examination of the blade. The examining background shall be dark and non-reflective. Blade shall be inspected on all surfaces: flat, on edges, and on ends.

4.4.3.2 Hardness. Hardness of the blades shall be determined by means of a Rockwell Hardness Tester having an N-Brale penetrator and a 30 kilogram load. Impressions shall be taken as close to the cutting edge as practicable but not on the beveled portion, and the average of three impressions per blade shall be considered the hardness value of the blade (see 3.1.2).

4.4.3.3 Stiffness. The stiffness of the blades shall be determined by use of a deflection test jig or similar device and as described in 4.4.3.3.2 and 4.4.3.3.3.

4.4.3.3.1 Testing device. The testing device shall be provided with two pairs of flat jaws, one pair of which is fixed to hold the blade securely in a horizontal position. The other pair of jaws shall be provided with suitable weights, scales, and arm so that when this pair of jaws is clamped to the blade which is held by the fixed pair of jaws, the deflection shall be indicated for a particular torque applied to the blade (see figure 9). The weight may be movable along the arm rather than in fixed position as shown in figure 9. Alternate testing equipment may be utilized, such as Tinius Olsen Torque type stiffness testing machine. Other types of testing equipment shall be approved by the procuring activity prior to use. When equipment other than that specified above, and illustrated in figure 9, is used, the maximum deflection limits specified in 3.6.4 above shall apply. All references to checking method for test jig specified herein shall not apply when the Tinius Olsen machine is used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.3.3.2 Deflection of point end. The blade shall be clamped in the fixed pair of jaws with the pointed end projecting, and the center line of the perforation perpendicular to the edge of the jaws. The perforation shall be completely within the grip of the jaws, with the end of the perforation nearest the point of the blade within 0.0312 inch of the edge of the jaws. The other pair of the jaws shall be clamped to the blade 0.25 ± 0.0156 inch from the fixed jaws. The applicable torque specified in table VI shall be applied to the blade through the movable jaws and the deflection noted to determine compliance with the requirements of 3.6.4.

Table VI. Torque Requirements

Size	Type	Torque
		Inch-pounds
1	AA	0.80
	BB	1.30
	CC	0.96
	DD	0.86
2	AA	1.48
	BB	1.73
	CC	1.86
	DD	1.48
	EE	0.96

4.4.3.3.3 Deflection of tang end. The pointed end of the blade shall be clamped in the fixed pair of jaws with the center line of the perforation perpendicular to the edge of the jaws, and with 0.375 ± 0.0156 inch of the perforation of size 1 blades and 0.4375 ± 0.0156 inch of the perforation of size 2 blade projecting beyond the edge of the jaws. The other pair of jaws shall be clamped to the blade 0.25 ± 0.0156 inch from the fixed jaws. A torque of 0.91 inch-pounds (0.80 inch-pounds for size 2, type EE blade) shall be applied to the blade through the moveable jaws and the deflection noted to determine compliance with the requirements of 3.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.3.4 Chemical analysis. At the request of the procuring agency, the contractor shall furnish millings or drillings of blade stock which shall be used in determining compliance with the carbon content requirement specified in 3.1.2. The contractor shall guarantee the sample millings or drillings to be of the same material as the stock from which the particular blade was manufactured. The millings or drillings shall weigh no less than four ounces, and shall be fine, clean, dry, and free from foreign matter. They shall be taken at slow speed and shall not be oxidized. Unless otherwise specified by the procuring agency, spectrographic analysis may be used in lieu of the chemical analysis test.

4.4.4 Sterility test. The sterility test shall be performed as described under Sterility Tests in the current edition of the U.S.P.

4.4.5 Freedom from pyrogens test. Place ten blades in a sterile, pyrogen-free flask. Pour 250 milliliters (ml.) of sterile, pyrogen-free saline test solution (U.S.P.) into the flask and then heat the contents of the flask at no less than 85 degrees Centigrade (° C.) for one hour. The resulting saline extract shall be submitted to the "Pyrogen Test" as described in the current edition of the U.S.P.

4.4.6 Safety tests. Cover 20 blades with 30 ml. of sterile, pyrogen-free saline test solution (U.S.P.) and warm at 37° C. for 24 hours. Decant the saline solution and test, undiluted, for both acute toxicity and tissue sensitivity. Determine acute toxicity by injecting 1.0 ml. of the eluate intravenously into each of 10 nonfasted, healthy, 28- to 32- gram mice. Observe the animals at 4, 24, and 72 hours after injection for any deaths or discernable symptoms of toxicity. Determine tissue reaction by injection 0.20 ml. of the eluate into each of ten sites on the back of a rabbit; the injections shall be made in such a manner as to raise a distinct bleb. Observe the injection sites at 4, 24, and 48 hours after injections for any evidence of redness, swelling or hardness. Trauma shall be ruled out with suitable saline control sites.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. The number of blades and handles, as specified in table VII, constitutes one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Table VII. Unit of Issue

Type	Size	FSN	Number of Blades	Number of Handles	Unit of Issue
EE	2	6515-299-8055	6		Pkg.
A	1	6515-344-7800		1	Each
A	2	6515-344-7820		1	Each
E	1	6515-344-7880		1	Each
D	1	6515-344-7920		1	Each
BB	2	6515-660-0007	6		Pkg.
AA	1	6515-660-0008	6		Pkg.
DD	1	6515-660-0009	6		Pkg.
CC	1	6515-660-0010	6		Pkg.
BB	1	6515-660-0011	6		Pkg.
AA	2	6515-660-0012	6		Pkg.
AA	1	6515-680-2086	144		Pkg.
DD	1	6515-680-2087	144		Pkg.
BB	1	6515-680-2088	144		Pkg.
AA	2	6515-680-2089	144		Pkg.
BB	2	6515-754-2833	144		Pkg.
CC	1	6515-821-2406	144		Pkg.
*	*	6520-286-4307	6	1	Pkg.

* Blade for federal stock number (FSN) 6520-286-4307 is type EE, size 2. Handle for FSN 6520-286-4307 is type D, size 2.

5.1.2 Packaging quantities. The number of units of issue indicated in table VIII shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VIII. Packaging Quantities

Type	Size	FSN	Unit package	Intermediate package	Exterior container
EE	2	6515-299-8055	1 unit	24 units	1,440 units
A	1	6515-344-7800	1 unit	12 units	576 units
A	2	6515-344-7820	1 unit	12 units	576 units
E	1	6515-344-7880	1 unit	12 units	576 units
D	1	6515-344-7920	1 unit	12 units	576 units
BB	2	6515-660-0007	1 unit	24 units	1,440 units
AA	1	6515-660-0008	1 unit	24 units	1,440 units
DD	1	6515-660-0009	1 unit	24 units	1,440 units
CC	1	6515-660-0010	1 unit	24 units	1,440 units

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Table VIII. Packaging Quantities (cont.)

Type	Size	FSN	Unit package	Intermediate package	Exterior container
BB	1	6515-660-0011	1 unit	24 units	1,440 units
AA	2	6515-660-0012	1 unit	24 units	1,440 units
AA	1	6515-680-2086	1 unit	12 units	72 units
DD	1	6515-680-2087	1 unit	12 units	72 units
BB	1	6515-680-2088	1 unit	12 units	72 units
AA	2	6515-680-2089	1 unit	12 units	72 units
BB	2	6515-754-2833	1 unit	12 units	72 units
CC	1	6515-821-2406	1 unit	12 units	72 units
*	*	6520-286-4307	1 unit	12 units	480 units

* Blade for FSN 6520-286-4307 is type EE, size 2. Handle for FSN 6520-286-4307 is type D, size 2.

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning. Each blade shall be cleaned by suitable methods to insure compliance with the requirements of paragraph 3.6.6.4.

5.1.3.2 Individual package.

5.1.3.2.1 Blade: composition A (carbon steel). Each blade shall be individually packaged in an envelope fabricated of aluminum foil with a heat sealable coating. The interior of the envelope shall be fitted with a strip of anticorrosive paper adhered to the inner surfaces of the aluminum foil in such a manner as to prevent contact of blade with inner surfaces of the envelope. The complete package shall be closed by heat sealing. One narrow edge of the envelope shall extend a minimum of 3/8 inch beyond the heat seal forming two flaps in such a manner as to permit easy removal of the contents by sterile techniques. The completed package shall be such as to maintain sterility until package is opened.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.3.2.2 Blade: composition B (stainless steel). Each blade shall be packaged in the same manner as specified in 5.1.3.2.1 except that the anticorrosive paper will not be required. The complete package shall be closed by heat sealing or pressure sealing. As an alternate, each blade shall be individually packaged in a combination paper-plastic envelope. One narrow edge of the envelope shall extend a minimum of 3/8 inch beyond the heat or pressure seal forming two flaps in such a manner as to permit easy removal of the contents by sterile techniques. The paper-plastic envelope shall be designed so as not to be resealable. As an alternate, the design of the paper-plastic envelope shall be such as to indicate that the envelope has previously been opened and resealed. The completed package shall be such as to maintain sterility until package is opened.

5.1.3.2.3 Handles. Each handle shall be packaged in accordance with MIL-P-116, method 1A-8.

5.1.3.3 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.4 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Individual package. Each individual package shall be marked with commercial name, size, lot (control) number, legend indicating that blade is sterile, and suitable instructions for opening. When individual package in closed by pressure seal, the legend "DO NOT RESEAL" shall be shown in red ink.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.3.2.2 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number, and date of this specification.
(b) Size and type of handles required (see 1.2.2).
(c) Size and type of blades required (see 1.2.3).
(d) Grade of handles required (see 3.1.1.2.1 and 3.1.1.2.2).
(e) For Defense Personnel Support Center procurement only. Unless otherwise specified, prior to delivery of any lots, samples shall be selected by the cognizant Government Quality Assurance Representative and forwarded to DPSC for inspection, to determine compliance with 3.6.3 Cutting edge and point, 3.6.3.1 Sharpness and cutting edge angle, 3.6.6.4 Freedom from corrosion and foreign matter, 4.4.3.1 Microscopic inspection. (see 4.1).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

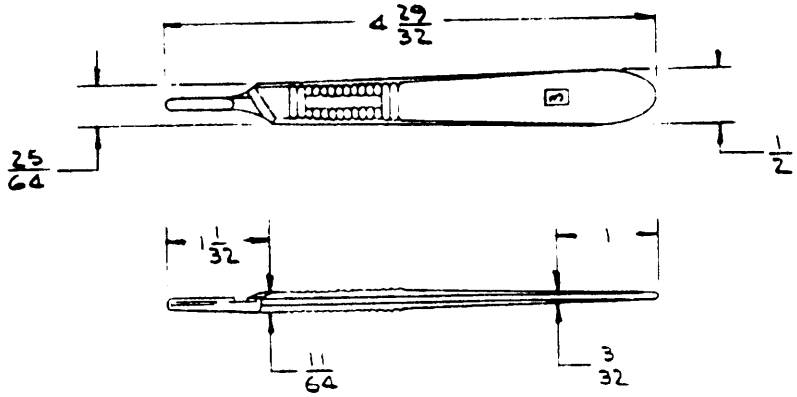


FIGURE 1a — SIZE 1 HANDLE NARROW NOSE; TYPE A, MINOR OPERATING, SHORT

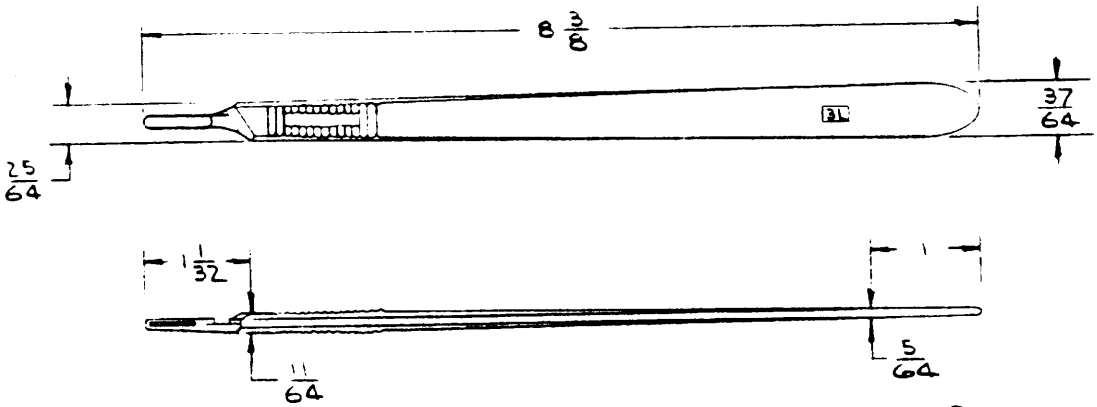


FIGURE 1b — SIZE 1 HANDLE, NARROW NOSE, TYPE B, MINOR OPERATING, LONG

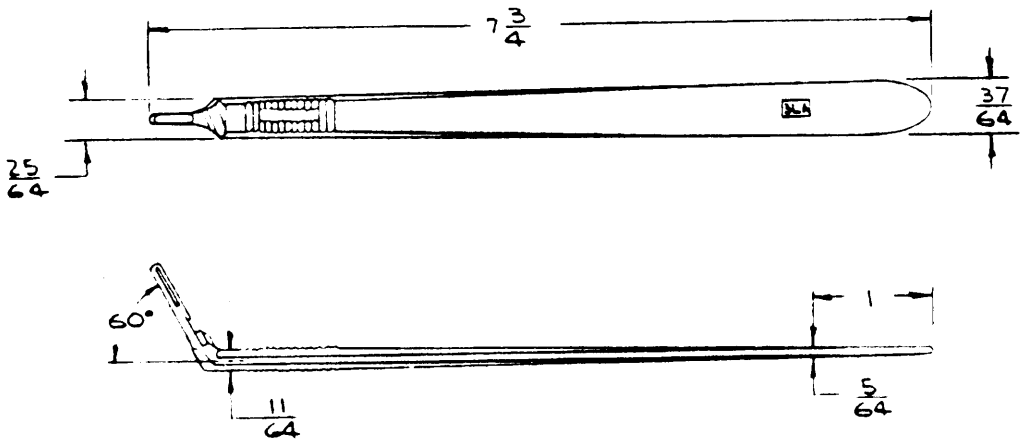


FIGURE 1c — SIZE 1 HANDLE, NARROW NOSE, TYPE C, MINOR OPERATING, OFFSET

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

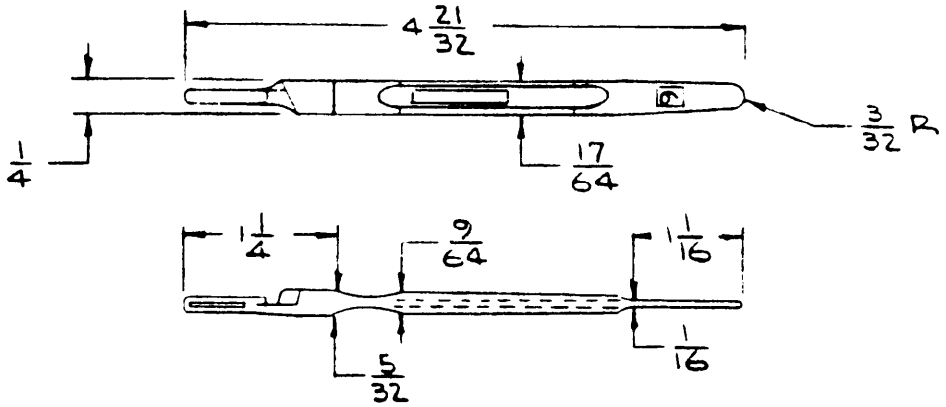


FIGURE 1d — SIZE 1 HANDLE, NARROW NOSE; TYPE D. DISSECTING, SHORT.

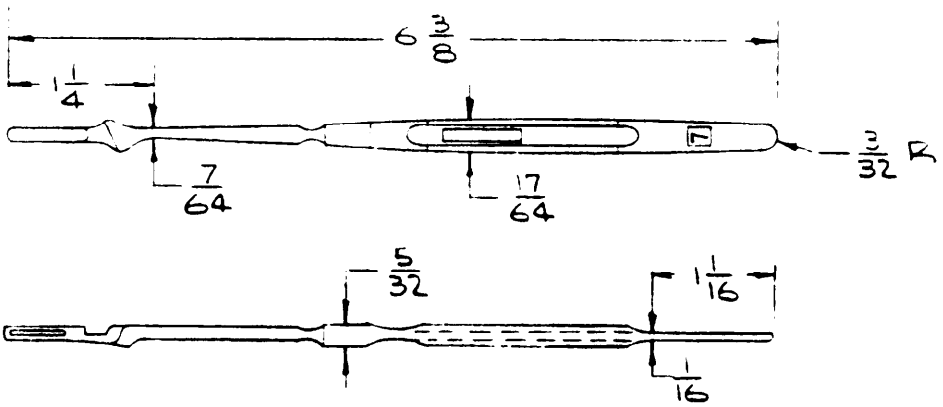


FIGURE 1e — SIZE 1 HANDLE, NARROW NOSE; TYPE E. DISSECTING, LONG.

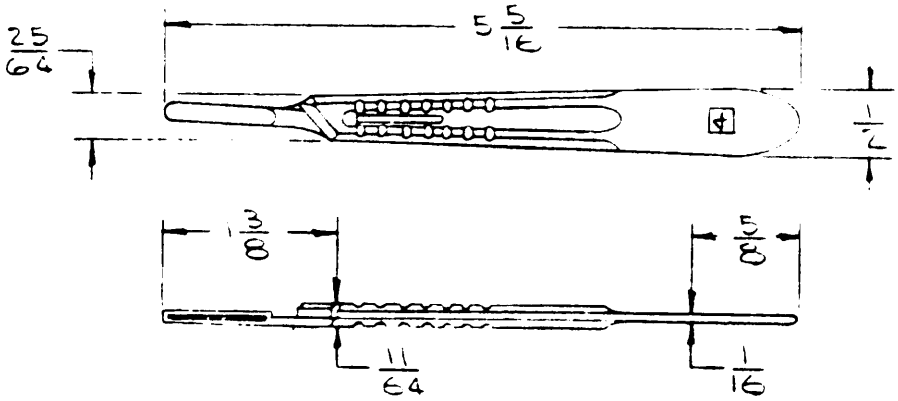


FIGURE 2a- SIZE 2 HANDLE, WIDE NOSE; TYPE A, MAJOR OPERATING, SHORT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

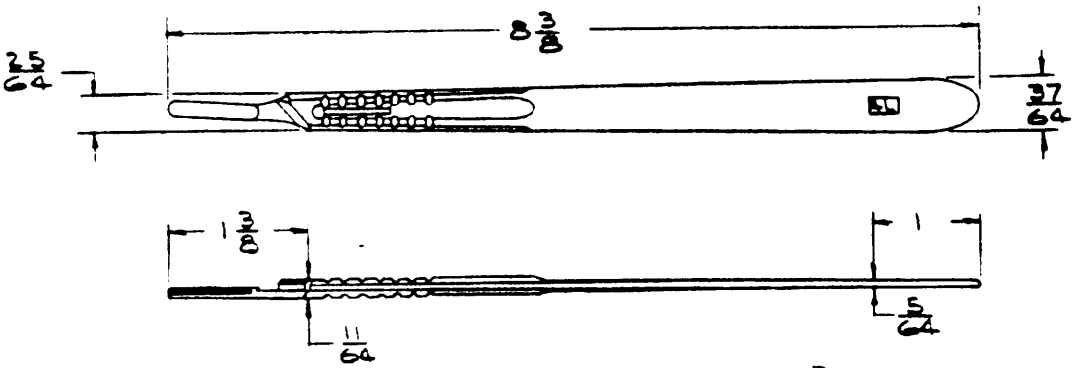


FIGURE 2b- SIZE 2 HANDLE, WIDE NOSE; TYPE B, MAJOR OPERATING, LONG.

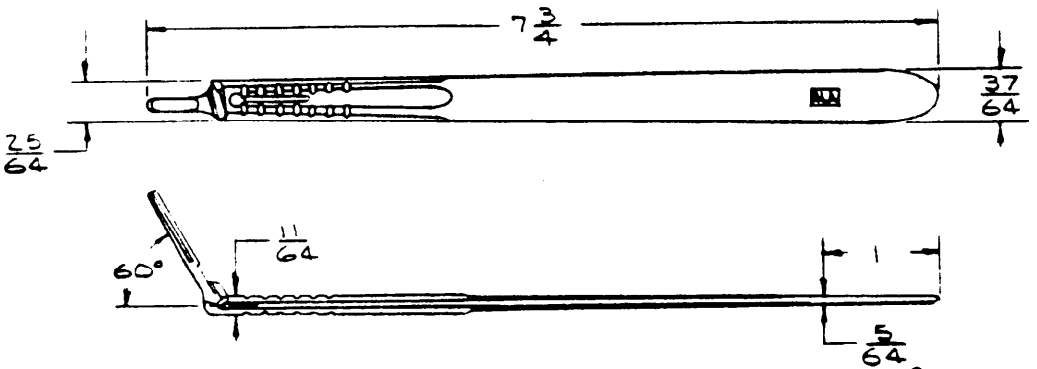


FIGURE 2c - SIZE 2 HANDLE, WIDE NOSE; TYPE C, MAJOR OPERATING, OFFSET.

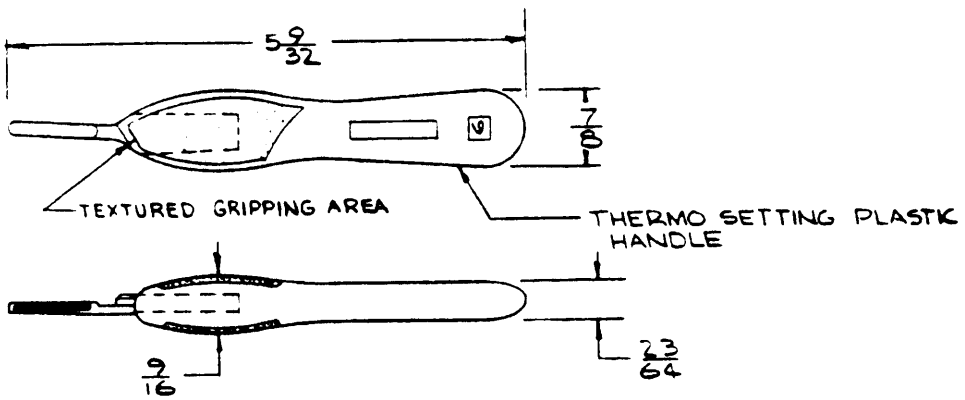
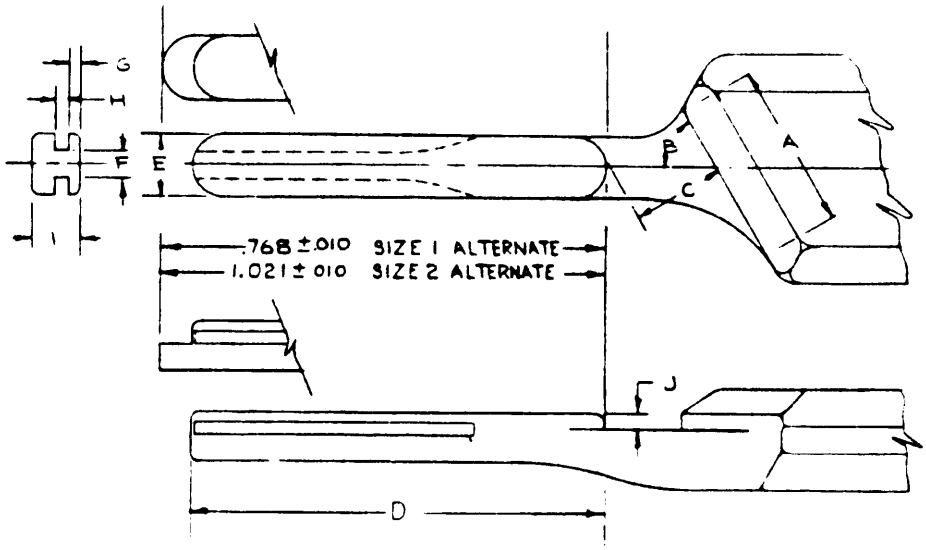


FIGURE 2d- SIZE 2 HANDLE, WIDE NOSE; TYPE D, DENTAL MECHANICS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.



	SIZE	A	B	C	D	E	F	G	H	I	J
HANDLES	1	0.240 TO .260	59 1/2 TO 60 1/2	.154 TC .162	.692 TO .698	.066 TC .094	.044 TO .046	0.014 TO .026	.016 TC .020	.082 TC .116	.020 TC .040
	2	.290 TC .320	54 1/2 TC 55 1/2	.237 TC .245	.943 TC .95	.136 TC .144	.0735 TC .0755	.014 TC .026	.0c TC .02C	.088 TC .12	.020 TC .040

DIMENSIONS SPECIFIED IN THE TABLE ARE IN INCHES EXCEPT B, WHICH IS IN DEGREES.

FIGURE 3 - DESIGN OF NOSE OF HANDLE

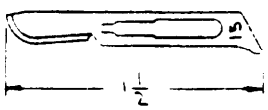


FIGURE 4a - SIZE 1 BLADE SMALL TANG, TYPE AA MINOR OPERATING, SMALL

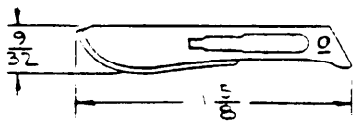
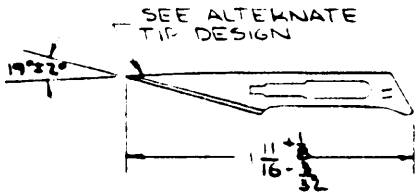


FIGURE 4b - SIZE 1 BLADE SMALL TANG, TYPE BB MINOR OPERATING, LARGE



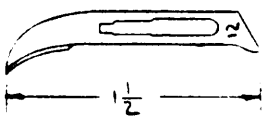
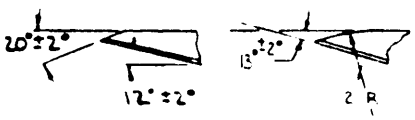


FIGURE 4d - SIZE 1 BLADE SMALL TANG, TYPE DD, BISTOURY, CURVED



ALTERNATE TIP DESIGNS

FIGURE 4c - SIZE 1 BLADE SMALL TANG, TYPE CC, BISTOURY, STRAIGHT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

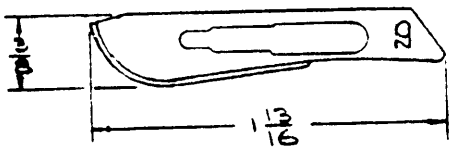


FIGURE 5a – SIZE 2 BLADE, LARGE TANG; TYPE AA, MAJOR OPERATING, SMALL.

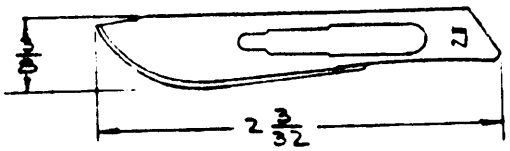


FIGURE 5b.– SIZE 2 BLADE, LARGE TANG; TYPE BB, MAJOR OPERATING, MEDIUM.

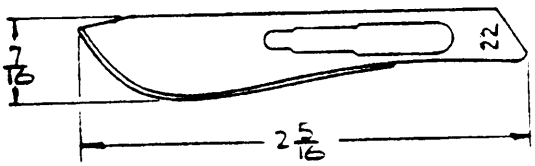


FIGURE 5c.– SIZE 2 BLADE, LARGE TANG; TYPE CC, MAJOR OPERATING, LARGE.

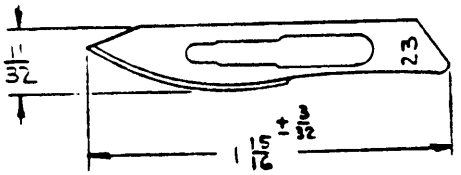


FIGURE 5d– SIZE 2 BLADE, LARGE TANG; TYPE DD, MAJOR OPERATING, SPECIAL.

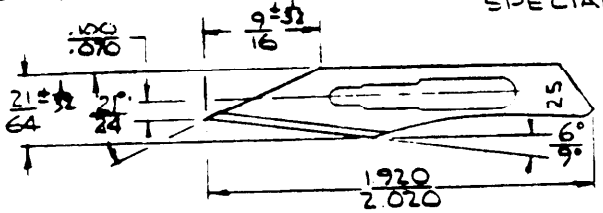
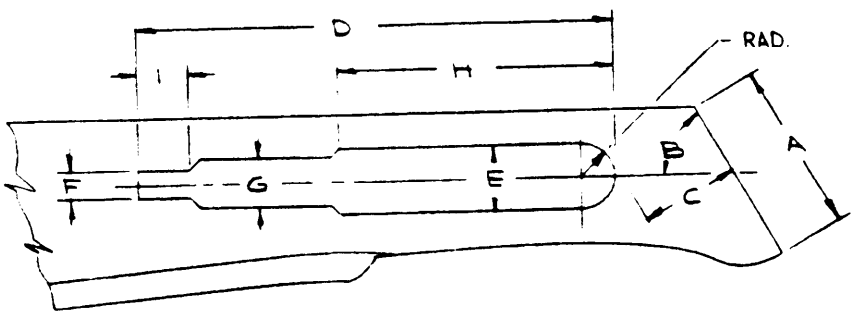


FIGURE 5e – SIZE 2 BLADE, LARGE TANG; TYPE EE, MAJOR OPERATING, BISTOURY, STRAIGHT.

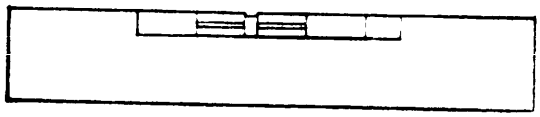


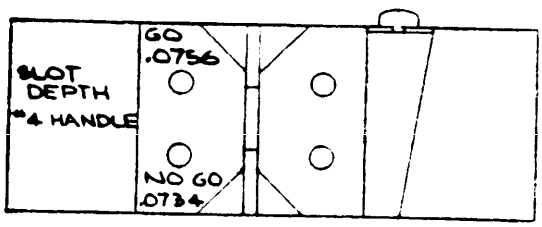
	SIZE	A	B	C	D	E	F	G	H	I
BLADES	1	0.240 TO .260	59 TO 61	0.140 TO .1518	0.7005 TO 0.7045	095 TO 100	.0462 TO .0483	.033 TO .071	0.390 TO .440	0.080 TO .110
	2	.275 TO .320	54 TO 56	.2252 TO .2348	0.9536 TO 0.9575	145 TO 150	.076 TO .0781	.090 TO .110	.420 TO .500	.110 TO .140

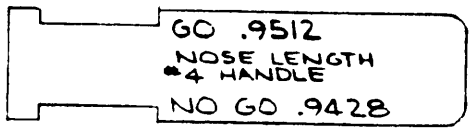
DIMENSIONS SPECIFIED IN THE TABLE ARE IN INCHES EXCEPT B WHICH IS IN DEGREES.

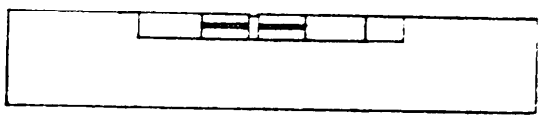
FIGURE 6.– DESIGN OF TANG OF BLADE.

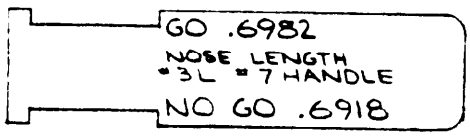
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.











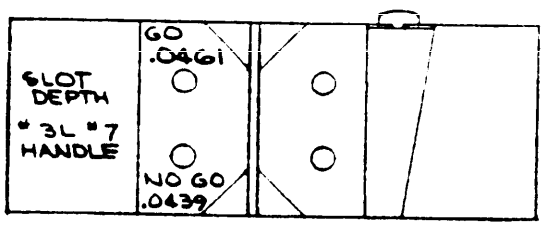


FIGURE 7.— "GO" AND "NO GO" GAGES FOR HANDLES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

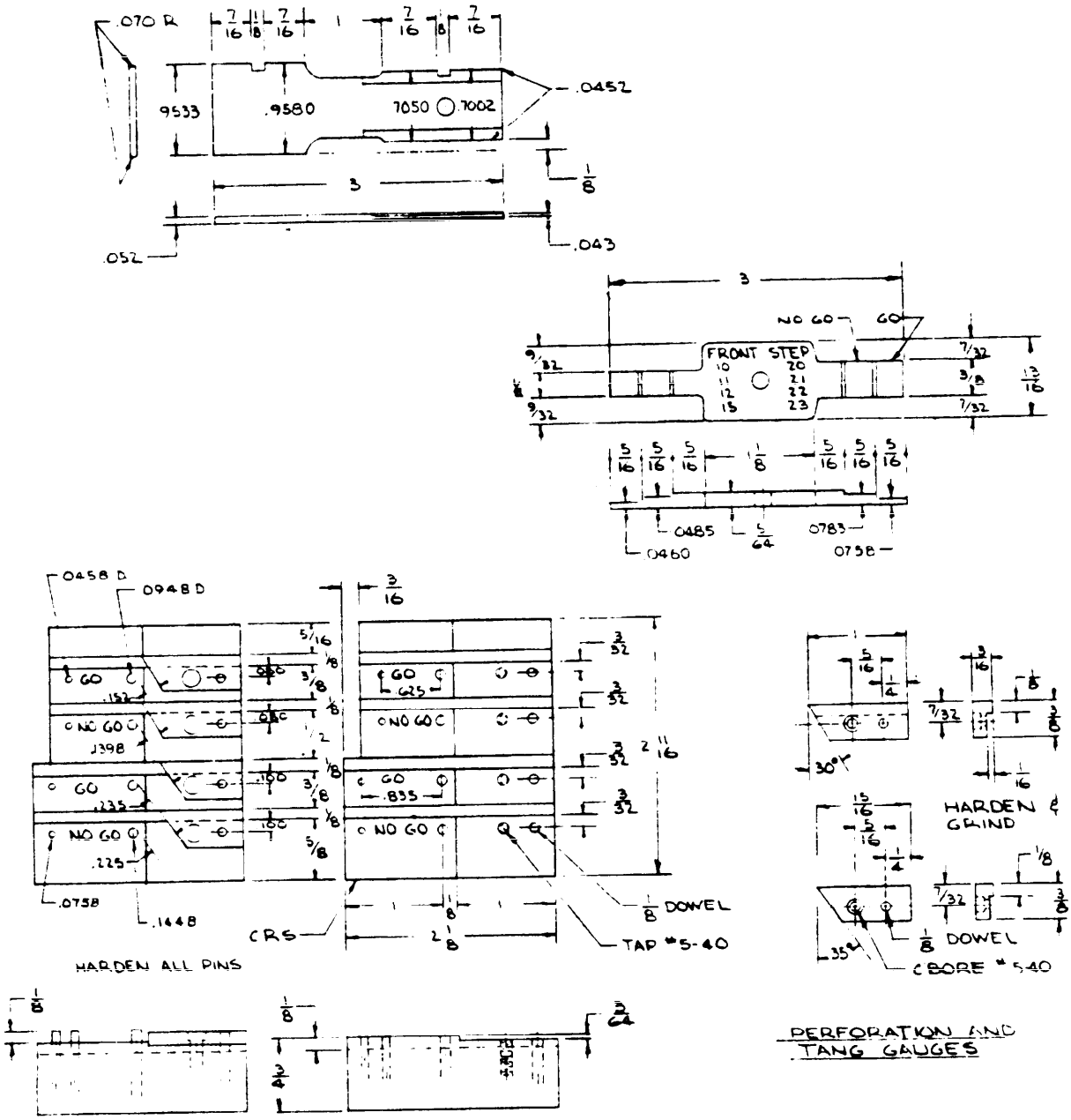


FIGURE 8 - "GO" AND "NO GO" GAGES FOR BLADES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

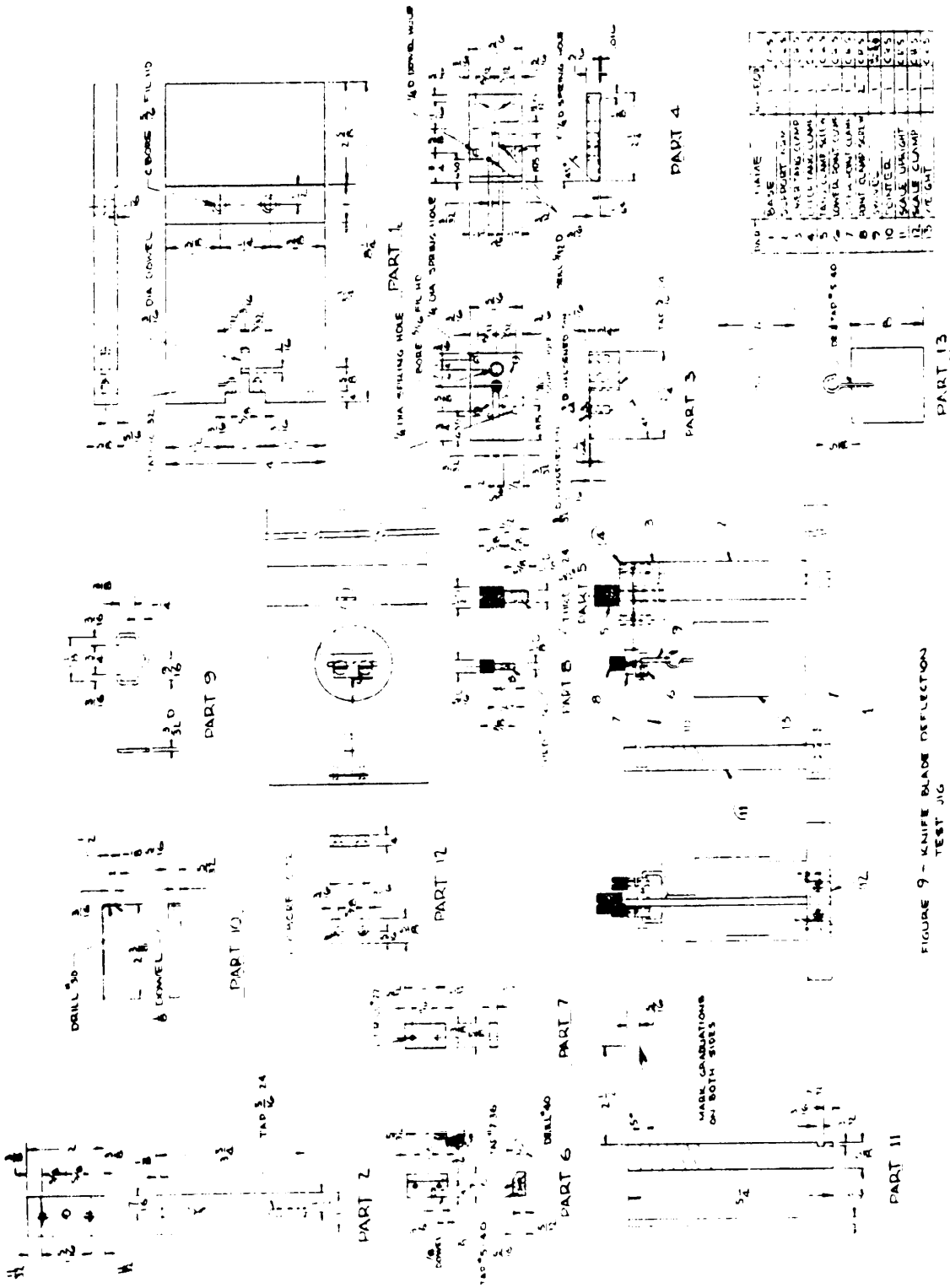


FIGURE 9 - KNIFE BLADE DEFLECTION TEST JIG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

GG-H-0080a(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

Fold

DPSC FORM 2418
AUG 80

GPO 303-223/Z 758

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.